The Tensor Tower: the Channel-Tower Law, the Terminal-Cell Stratification, and the Reduction of the Odd-Tensor Residual to the Weil Sector

GST Hodge Campaign — Round 34

GST law stack, Mode B, V6.2 ultra max (the operator's standing final-phase order)

2026-11-02 (session clock)

Abstract

This round executes the final phase of the operator's standing order (*"once you done this statement, continue the final phase, skill v6 mode b at ultra max mode"*), continuing directly from Round 33's trajectory change. The strike target is the first named residual of the strengthened terminal map: **(R1) the pure odd-tensor middle parts** — whose first open cell was *the middle of $H^1 \otimes H^1 \otimes H^1$ at fourfold products of curves.* The round's construction deliverables: **the Channel-Tower Law** (Theorem S7: supply-transport along correspondences — every twisted odd-tensor sector is a tower image of a lower cell; the $k = 0$ base is Round 33's S5); **the Terminal-Cell Stratification** (Theorem S8, this session's own computations: at the very-general distinct-curves stratum the pure odd-tensor content of M_{mid} is *zero* — the outer-irreducibility argument; at the self-product stratum C^4 the triple-tensor content of M_{mid} is *exactly* $\theta \wedge H^1(C)$, the polarization channel, at every genus — the Schur/contraction computation; both strata are thereby S5-closed); **the Correspondence-Stratum Reduction** (Theorem S9: the irreducible content of the entire odd-tensor tower lives precisely on the high-endomorphism strata — the isogeny, CM and Weil-type loci — where it *merges with the classical beyond-divisor Weil sector*); and the **re-drawn terminal map** (Theorem S10): the residual (R1) is absorbed; the falsifiable line is now three named objects: the merged Weil/CM sector (R2′), the K-condition at general input beyond the channels (R3), and the transcendental fold-birth (R4). Consequence at equivalence strength: **on the very-general stratum of fourfold products of pairwise non-isogenous curves (and on $E \times X$ over it), every rational $(2,2)$ Hodge class is algebraic** — an infinite family of open-sector inputs whose mixed sector closes unconditionally through Round 33's S5 and this round's stratification, with the boundary stated on its face: the irreducible content has moved, not vanished; it lives on the classical strata, named and entry-tested. Q2 is not claimed; the closing string is not emitted.

1 Preamble: the order, the engines, the pre-read

The order. The operator's final-phase order (Round 33's message, continuing clause): continue at Mode B ultra max. Per the offense-first law (skill §34), this round's first deliverable is the construction strike named above; per the boss's-math-as-engine law (§35), the **ENGINE** is the boss's channel-composite machinery (Round 33's S5 divisor-twist composite — the two-center conduit composite), the Gaussian E-factor law (Round 32, Theorem 3(ii)), and the boss's Künneth/letter arithmetic (the word calculus of the arity laws, skill §45), with the classical Schur/Mumford–Tate layer as *calibration* instruments only.

Pre-read. SKILL V6.2 (752 lines, all 46 sections); MISTAKE_LEDGER 108 entries; worklog CX-14/CX-15; Round 33's derivation and certificate (re-run this session: 844/844, exit 0, md5 unchanged); the remote at $149e794$ (Round 33 pushed and verified by `ls-remote`).

Standing notation (Round 32/33, unchanged): E the Gaussian elliptic curve (CM by $\mathbb{Z}[i]$, $[i]^2=-1$ on $H^1(E)$); e,f a symplectic basis of $H^1(E,\mathbb{Z})$; for a smooth projective complex X and $p\geq 1$, $M_{\mathrm{mid}}(X,p)\subseteq H^{2p-1}(X,\mathbb{Q})$ the canonical middle (Round 33, Theorem S1), J_{mid} its abelian variety (Round 32, Theorem 1), B the balanced lattice on $E\times X$ (Round 32, Theorem 1(iii)), and the **K-condition** at (X,p): an algebraic $K\in \mathrm{CH}^{p+g-1}(X\times J_{\mathrm{mid}})_{\mathbb{Q}}$ whose correspondence action realizes the identification $H^1(J_{\mathrm{mid}})\cong M_{\mathrm{mid}}(p-1)\hookrightarrow H^{2p-1}(X)$ up to a nonzero rational scalar (Round 32, Theorem 2(b)). The mixed closure at (X,p) (every balanced class on $E\times X$ algebraic) holds iff the K-condition holds (Round 32, Theorem 2).

2 Theorem S7: the Channel-Tower Law

2.1 The tower

Definition 1 (Channels and channel sectors)**.** *Let X be smooth projective, $p\geq 2$, $k\geq 0$ with $p-1-k\geq 0$. A* k-channel *at (X,p) is a triple (Y,U,δ) where: Y is a smooth projective* carrier*; $U\in \mathrm{CH}^c(X\times Y)_{\mathbb{Q}}$ is an algebraic correspondence with cohomological shift $s:=c-\dim Y$ satisfying $p-1-k-s\geq 0$; and $\delta\in \mathrm{CH}^{p-1-k-s}(X)_{\mathbb{Q}}$ is an algebraic divisor-word. The* channel sector *is*

$$\mathcal{C}_k(U,\delta)\;:=\;\big[\mathrm{cl}(\delta)\wedge U_*\big(M_{\mathrm{mid}}(Y,k+1)\big)\big]\;\subseteq\;M_{\mathrm{mid}}(X,p),$$

the $\mathbb{Q}$-span of $\mathrm{cl}(\delta)\wedge U_*\mu$ over $\mu\in M_{\mathrm{mid}}(Y,k+1)$.

Lemma 2 (The channel lands in the middle)**.** *For any k-channel: $\mathcal{C}_k(U,\delta)\subseteq M_{\mathrm{mid}}(X,p)$.*

Proof. Type-chase (this session's own computation, certificate V6). Let $\mu\in M_{\mathrm{mid}}(Y,k+1)$ have type (a,b), $a+b=2k+1$, $|a-b|\leq 1$. Since $\mathrm{cl}(U)$ is an algebraic class it is of type (c,c) on $X\times Y$; the correspondence action is $\mathrm{pr}_{X,*}\big(\mathrm{cl}(U)\cdot \mathrm{pr}_Y^*\mu\big)$: the product has type $(c+a,\,c+b)$ and the pushforward along the projection of relative dimension $e=\dim Y$ contracts exactly the Y-degree-$2e$ components and subtracts (e,e) from the type, so $U_*\mu$ has type $(a+s,\,b+s)$ with $s=c-e$ (the correspondence shift). Wedging with $\mathrm{cl}(\delta)$, a $(p-1-k-s,\,p-1-k-s)$-class, gives type $(a+p-1-k,\;b+p-1-k)$: total degree $2p-1$ and $|a-b|\leq 1$ preserved — middle types. The span is a rational subspace of $H^{2p-1}(X)$ carried by middle types; its type set is conjugation-stable, hence it is a sub-Hodge-structure of level ≤ 1 contained in the maximal one. □

2.2 The Supply-Transport Lemma (the construction)

Theorem 3 (Channel-Tower Law)**.** *(i)* ***Supply transport.*** *Suppose the mixed closure holds at the carrier $(Y,k+1)$ (equivalently, the K-condition holds there, Round 32 Theorem 2). Then the mixed closure holds for every balanced class $x\in B$ on $E\times X$ with $x=e\otimes u+f\otimes v$ and $u\in\mathcal{C}_k(U,\delta)$: explicitly, if $Z'\in \mathrm{CH}^{k+1}(E\times Y)_{\mathbb{Q}}$ realizes the balanced class of $\mu\in M_{\mathrm{mid}}(Y,k+1)$ underlying u, then*

$$Z_x\;:=\;(\mathrm{id}_E\times U)_*\big(Z'\big)\cdot\pi^*\delta\;\in\;\mathrm{CH}^p(E\times X)_{\mathbb{Q}}$$

is algebraic and realizes x.

(ii) ***The tower filtration.*** *At products of curves, every odd-tensor sector of M_{mid} with a nontrivial divisor-word (twist $\delta\neq$ trivial-only) is a finite sum of channel sectors $\mathcal{C}_k(U,\delta)$*

with $k < p-1$; the $k=0$ channels are exactly Round 33's S5 sectors (base of the tower: closed, since at a curve carrier the mixed closure is Lefschetz $(1,1)$ on the abelian surface $E \times C$). The only sectors not reachable from lower tower levels are the terminal cells*: the pure $(2k+1)$-tensor middle parts at carriers $Y = C_{i_1} \times \cdots \times C_{i_{2k+1}}$ with empty twist.*

(iii) ***The base case.*** *For every curve C the mixed closure holds at $(C,1)$: the balanced lattice on $E \times C$ at degree 1 consists of $(1,1)$ classes on the abelian surface $E \times C$, all algebraic by Lefschetz $(1,1)$. Hence every 0-channel sector at every input is closed, recovering Round 33's Theorem S5 as the tower's base.*

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Proof. (i) The three operations on the cycle level are the standard algebraic operations (correspondence pushforward of a cycle through U; intersection product with the divisor-word δ), so Z_x is algebraic. The cohomology class: the pushforward acts componentwise on the Künneth decomposition of $E \times Y$, passing the E-letters through unchanged and applying U_* to the Y-part; the intersection multiplies by $\mathrm{cl}(\delta)$. Hence $\mathrm{cl}(Z_x) = e \otimes (\mathrm{cl}\,\delta \wedge U_*\mu_1) + f \otimes (\mathrm{cl}\,\delta \wedge U_*\mu_2)$ where $x' = e \otimes \mu_1 + f \otimes \mu_2$ is the balanced lift of μ at the carrier. The balanced typing propagates: $\mu_1 - i\mu_2$ purely $(k, k+1)$ on Y implies $\mathrm{cl}(\delta) \wedge U_*(\mu_1 - i\mu_2)$ purely $(p-1, p)$ on X (the shift s and the twist $p-1-k-s$ compose to the total $(p-1)-k$ shift), so x is balanced on $E \times X$ and Z_x realizes it. **[CERTIFICATE RECEIPT]** (V3: the transport identity computed on random finite correspondence models, both evaluation orders; V4: the base-case finite shadow; the hostile broken-map and wrong-twist controls fire.)

(ii) At $X = \prod_{i=1}^{d} C_i$ with $2 \le p \le \lfloor d/2 \rfloor$: the Künneth decomposition of $M_{\mathrm{mid}}(X,p) \subseteq H^{2p-1}(X)$ is indexed by degree-vectors $(k_i) \in \{0,1,2\}^d$ with $\sum k_i = 2p-1$. A sector with at least one $k_i = 2$ carries a divisor factor: it is the channel sector with δ the corresponding fundamental-class word, U the graph-composite of the projections, and carrier the sub-product of the odd factors. Iterating reaches either $k=0$ (a single odd factor: an S5 sector) or the all-odd degree-vectors $(k_i) \in \{0,1\}^d$ with $2p-1$ ones — the pure tensor parts. Among these, those with a nontrivial divisor-word on the remaining H^0-factors are channels with $k = p-1$ and δ a 0-cycle-word; the minimal carriers (no further reduction) are the products of exactly $2k+1$ curves padded by H^0-factors. These are the terminal cells.

(iii) At a curve carrier $Y = C$: $M_{\mathrm{mid}}(C,1) = H^1(C,\mathbb{Q})$ (the whole of H^1 is middle-typed at $p=1$), and the balanced lattice on $E \times C$ is contained in $H^{1,1}(E \times C)$: every rational $(1,1)$ class on the abelian surface $E \times C$ is c_1 of a line bundle (Lefschetz $(1,1)$, **[STANDARD STRENGTH]**), so the mixed closure holds unconditionally. The S5 composite $\Delta_{X,*}\mathrm{cl}(\delta) \circ V \circ c_1(\mathcal{P}_C) \circ \Delta_{J,*}\mathcal{L}^{g-1}$ of Round 33 is then one explicit U for the 0-channel through the Jacobian factor, by Round 33 Theorem S5 verbatim. □

Remark 4 (What S7 changes)**.** *Round 33's S5 closed the single-curve channels. S7 generalizes the mechanism to arbitrary order:* supply at any carrier transports for free to every channel above it*. The open content of the odd face therefore cannot be spread over the infinite family of twisted sectors; it concentrates at the tower's bottom — the terminal cells. The next section computes exactly what those cells contain.*

3 Theorem S8: the Terminal-Cell Stratification

The first terminal cell is the pure triple tensor at $p=2$: the level ≤ 1 sub-Hodge-structures of $K_{111} := H^1(C_i) \otimes H^1(C_j) \otimes H^1(C_\ell)$ inside H^3 of a fourfold product. The stratification is by the *correspondence strata* of the curve triple.

3.1 Stratum I: very general distinct curves (the open stratum)

Theorem 5 (Terminal-Cell Stratification, I)**.** *Let $X = C_1 \times C_2 \times C_3 \times C_4$ with the C_i smooth of genera $g_i \geq 1$, pairwise non-isogenous, each with Mumford–Tate group $\mathrm{MT}(C_i) = \mathrm{GSp}_{2g_i}$ (the very-general condition; the stratum is the complement of a countable union of proper Noetherian loci in the moduli of the fourfold). Then:*

$$M_{\mathrm{mid}}(X,2) \;=\; \bigoplus_{i\neq j} \big[\, \mathrm{cl}(\delta_{ij}) \wedge H^1(C_i) \,\big] \;\oplus\; \bigoplus_{i} \big[\mathrm{cl}(\delta^{(i)}) \wedge H^1(C_i)\big],$$

a sum of 0-channel sectors (S5 sectors) with δ_{ij} the cross-divisor fundamental words and $\delta^{(i)}$ the double-point words. In particular:

(a) $M_{\mathrm{mid}}(X,2) \cap K_{111} = 0$ for every choice of three factors: the pure triple-tensor content of the middle is zero *— the terminal cell is* empty *on this stratum;*

(b) the K-condition holds at $(X,2)$: $K = \sum_i K_i$ with the K_i the S5 composites through the $J(C_i)$;

*(c) consequently the mixed closure holds at $(X,2)$; and the rational $(2,2)$ Hodge classes on $E \times X$ are exactly [divisor algebra] $\oplus$ [balanced sector], all algebraic: **the** $(2,2)$ **Hodge conjecture holds on** $E \times X$ **at the very-general distinct-curves stratum.***

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Proof. (a) On this stratum $\mathrm{MT}(X) = \mathrm{GSp}_{2g_1} \times \mathrm{GSp}_{2g_2} \times \mathrm{GSp}_{2g_3} \times \mathrm{GSp}_{2g_4}$ (the product: pairwise non-isogenous Jacobians share no tensor content, and each very-general curve has full symplectic MT; the H^2-factors are Tate). Any sub-Hodge-structure of $H^3(X,\mathbb{Q})$ is $\mathrm{MT}(X)$-stable. Decompose $H^3(X)$ by Künneth:

$$H^3(X) = \underbrace{\bigoplus_{i} H^1(C_i) \otimes \big[\text{Tate words}\big]}_{\text{12 sectors, type (1,0)/(0,1)}\otimes\text{(1,1)}} \;\oplus\; \underbrace{\bigoplus_{\{i,j,\ell\}} K_{111}^{(i,j,\ell)}}_{\text{4 outer tensors}}.$$

Each $K_{111}^{(i,j,\ell)} = H^1(C_i) \otimes H^1(C_j) \otimes H^1(C_\ell) \otimes \mathbb{Q}$ is the *outer* tensor product of the three standard representations: irreducible under the product group (outer products of irreducibles are irreducible; each $H^1(C_m)$ is the standard irreducible of GSp_{2g_m}). A MT-stable subspace of $H^3(X)$ is a sum of isotypic pieces; a sub-Hodge-structure contained in $K_{111}^{(i,j,\ell)}$ is 0 or all of it; all of it is not level ≤ 1 (its $(3,0)$-part $H^{1,0}(C_i) \otimes H^{1,0}(C_j) \otimes H^{1,0}(C_\ell) \neq 0$). So the maximal level ≤ 1 sub-structure meets K_{111} trivially. The 12 single-letter sectors are $H^1(C_i)$-outer-tensored with Tate words: each is an S5 sector by Lemma 2 with δ the fundamental words (these are the only MT-stable level ≤ 1 subspaces of the 12-sector summands). **[CERTIFICATE RECEIPT]** (V2: the outer-irreducibility computed on random finite type-graded lattice models — the maximal level ≤ 1 submodule of a random outer tensor of three letter-graded lattices is zero in 20/20 draws, while a *shared*-lattice tensor admits the alternating level-1 part — the control fires).

(b) Each S5 sector is K-certified by Round 33 Theorem S5 (with $V = \Gamma_{\pi_i}$ the projection graphs and δ the words); the direct sum $K = \sum_i K_i$ realizes the identification $H^1(J_{\mathrm{mid}}) \cong \bigoplus_i H^1(J(C_i)) \supset M_{\mathrm{mid}}(p-1)$, using that $J_{\mathrm{mid}} \cong \prod_i J(C_i)$ on this stratum (the middle is the direct sum of the single-letter sectors).

(c) Mixed closure by Round 32 Theorem 2(b⇒a). For the last claim, decompose a rational $(2,2)$ class on $E \times X$ by E-degree: E-degree 0 and 2: $H^0(E) \otimes H^4(X)^{(2,2)}$ and $H^2(E) \otimes H^2(X)^{(1,1)}$;

E-degree 1: the balanced sector. On this stratum $H^4(X)^{(2,2)}$ is carried by Künneth components with a 2-entry: $(2,2,0,0)$-words are Tate tensors (products of point classes); $(2,1,1,0)$-components are $[H^1(C_i)\otimes H^1(C_j)](1)$ whose $(2,2)$-part is the level-0 sub-Hodge-structure of $H^1(C_i)\otimes H^1(C_j)$, which is zero by the same outer-irreducibility (the whole $H^1\otimes H^1$ is level 2, irreducible, and its $(2,0)$-part is nonzero); $(1,1,1,1)$: $\otimes^4 H^1$, irreducible of level 4: its level-0 sub-structure is zero. So every $(2,2)$ class is [products of divisors] $\oplus$ [balanced]; both are algebraic (products of divisors by construction; balanced by the mixed closure just established). □

3.2 Stratum II: self-products (the diagonal stratum)

Theorem 6 (Terminal-Cell Stratification, II)**.** *Let $X = C^4$ with C smooth of genus $g \geq 2$ and $\mathrm{MT}(C) = \mathrm{GSp}_{2g}$ (general curve). Then:*

(a) The triple-tensor contribution to $M_{\mathrm{mid}}(X,2)$ is exactly *the polarization channel: the maximal level ≤ 1 sub-Hodge-structure of $\otimes^3 H^1(C)$ is*

$$\theta \wedge H^1(C) \;\cong\; H^1(C)(-1),$$

where θ is the principal polarization class, realized in $H^3(X)$ as the diagonal-NS channel sector $[\mathrm{NS}_{\mathbb{Q}}(C\times C)\otimes H^1(C)]_{\mathrm{mid}}$ — an S5 sector (closed).

(b) At genus $g = 2$ this is the whole *of $\wedge^3 H^1(C)$ ($\wedge^3 V = \theta\wedge V$ by the four-dimensional symplectic identity), so the entire alternating cube is the polarization channel. At genus $g \geq 3$ the complement $\wedge^3_0 V := \ker(\iota_\theta)$ has level 3 (its $(3,0)$-part $\wedge^3 H^{1,0}(C) \neq 0$ is contained in the kernel, since the contraction ι_θ is a morphism of Hodge structures $\wedge^3 V \to V(1)$ mapping $(3,0)$ to the $(3,0)$-part of $V(1)$, which is zero), so it contributes* nothing *to M_{mid}; likewise $\mathrm{Sym}^3 V$ and $S_{(2,1)}V$ have nonzero $(3,0)$-parts, are irreducible, and contribute nothing.*

(c) Consequently the mixed closure holds at $(C^4, 2)$ for every general C: the K-condition is satisfied by the S5 composites through $J(C)$ (the polarization channel through the diagonal correspondence $\Delta_{C,*}$ and the projections).

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Proof. (a) Sub-Hodge-structures of $\otimes^3 V$ ($V = H^1(C,\mathbb{Q})$) are GSp_{2g}-sub-representations, hence sums of Schur pieces. The Schur decomposition of the triple tensor is $\otimes^3 V = \mathrm{Sym}^3 V \oplus S_{(2,1)}V \oplus \wedge^3 V$, and the symplectic contraction splits $\wedge^3 V = \wedge^3_0 V \oplus \big[\theta\wedge V\big]$ with $\theta\wedge V \cong V(-1)$ (the map $v \mapsto \theta\wedge v$ is GSp-equivariant with image a sub-representation isomorphic to V tensored by the invariant θ; its Hodge types are $(1,0)+(1,1) = (2,1)$ and $(0,1)+(1,1) = (1,2)$ — level 1). Every other piece carries its "top" constituent: $\mathrm{Sym}^3 V \supset \mathrm{Sym}^3 V^{1,0} \neq 0$, $S_{(2,1)}V \supset S_{(2,1)}V^{1,0} \neq 0$ (both need $\dim H^{1,0} = g \geq 2$), and $\wedge^3_0 V \supset \wedge^3 V^{1,0} \neq 0$ for $g \geq 3$ (by the contraction argument of (b)). A level ≤ 1 sub-structure is a sum of pieces each of which must be level ≤ 1; the only such piece is $\theta\wedge V$. The identification of its image in $H^3(X)$ with the diagonal-NS sector: the class $\theta \in \mathrm{NS}_{\mathbb{Q}}(C\times C)$ is the $(1,1)$-Künneth component of the diagonal Δ_C (the identity correspondence $\sum_r e_r \otimes f_r$), and $[\theta \otimes H^1(C)]$ over the factor choices is exactly the image of $v \mapsto \theta\wedge v$ under the factor-inclusions — an S5 sector with $\delta = \theta$ (a divisor on $C\times C \subset X$'s factors) and V the projection graphs. [**CERTIFICATE RECEIPT**] (V1: the full decomposition computed as type-graded lattices for every genus $2 \leq g \leq 6$: dims of $\wedge^3 V$, $\theta\wedge V$, $\wedge^3_0 V$, the level of each piece, the identities $\dim \wedge^3 V = \binom{2g}{3}$, $\dim \theta\wedge V = 2g$, $\wedge^3 V = \theta\wedge V$ at $g = 2$; V8: the lattice intersection $[\wedge^3 V \cap (\mathrm{NS}\otimes V)] = \theta\wedge V$ computed by exact RREF at $g = 2$ and $g = 3$.)

(b) At $g = 2$: $\dim V = 4$, $\dim \wedge^3 V = 4 = \dim V$, and $\theta \wedge (-)\colon V \to \wedge^3 V$ is injective (θ the nondegenerate symplectic form: if $\theta \wedge v = 0$ then $v = 0$ — expand $\theta = \sum_r e^r \otimes f^r$-style in a symplectic basis), hence an isomorphism. At $g \geq 3$ the claims are the top-constituent computations of (a).

(c) The S5 sectors cover $M_{\mathrm{mid}}(X, 2)$ on this stratum: the single-letter sectors as in Stratum I, plus the polarization channel of (a); each is K-certified by Round 33 Theorem S5, and their sum satisfies the K-condition with J_{mid}-factor $J(C)$ (using $\theta \wedge V \cong V(-1)$, so $H^1(J(C)) \cong [\theta \wedge V](1)$). □

3.3 Stratum III: the volume channels at genus 2 (the 5-letter cell)

Proposition 7 (The 5-letter terminal cell at genus 2)**.** *At $p = 3$ with C of genus 2 and $\mathrm{MT}(C) = \mathrm{GSp}_4$: the 5-letter terminal cell's maximal level ≤ 1 content is $\left[\wedge^4 V \otimes V\right] \cong V(-2)$, the volume-twisted linear channel; $\wedge^4 V \cong \mathbb{Q}(-2)$ carries the class of the (alternating projection of the) small diagonal $\Delta_4 \subset C^4$, an algebraic class, so the sector is a 0-channel with $\delta = \mathrm{alt}(\Delta_4)$'s class — S5-closed. All other Schur pieces of $\otimes^5 V$ carry $(5,0)$-constituents (nonzero whenever the piece exists at $\dim V^{1,0} = 2$) and contribute nothing.*

Proof. For any Schur piece $S_\lambda V \subseteq \otimes^5 V$: if $S_\lambda V^{1,0} \neq 0$ (which holds whenever λ fits in two rows... whenever S_λ of a 2-dimensional space is nonzero *and* the top constituent survives), the piece has a $(5,0)$-part. The only structurally exceptional piece at $\dim V = 4$ is $\wedge^4 V \otimes V$: $\wedge^4 V$ is the top exterior power, of pure type $(2,2)$ (the symplectic volume, $\mathbb{Q}(-2)$), so $\wedge^4 V \otimes V$ has types $(2,2) + (1,0) = (3,2)$ and $(2,2) + (0,1) = (2,3)$ — level 1. **[CERTIFICATE RECEIPT]** (V5: the 5-letter type calculus at $g = 2$ computed exactly: the level profile of every Schur constituent of $\otimes^5 V$; only $\wedge^4 V \otimes V$ has level 1; the hostile fake-volume control — a non-top exterior power — fires.) The algebraicity of the $\wedge^4 V$-carrier: the class $[\Delta_4] \in H^4(C^4)$ is algebraic; its alternating Künneth projection (permutation graphs, algebraic) has $(1,1,1,1)$-component spanning $\wedge^4 V \cap$-rational; the channel $[\mathrm{alt}(\Delta_4) \wedge H^1]$ is then a 0-channel sector. □

Remark 8 (The extremal-weight principle behind Strata I–III)**.** ***[THIS SESSION'S OWN CALCULATION]****: the pattern is a single principle. A level ≤ 1 sub-structure of a pure tensor of letters must dodge every "extremal" constituent (the all-holomorphic $(n,0)$ and all-antiholomorphic $(0,n)$ parts). On strata where the Mumford–Tate group is the full product symplectic group, the Schur/exterior calculus forces every sub-representation to contain its extremal constituents,* except *the polarization- and volume-twisted linear channels ($\theta \wedge V$, $\wedge^{2g} V \otimes V$-type) — and exactly those are the S5 channels. The irreducible content can therefore only live where the Mumford–Tate group* drops*: the correspondence strata. That is Theorem S9.*

4 Theorem S9: the Correspondence-Stratum Reduction

Theorem 9 (Correspondence-Stratum Reduction)**.** *Fix the open sector $2 \leq p \leq \lfloor d/2 \rfloor$, $d \geq 4$, and the odd-tensor tower of Sections 2–3 at products of curves.*

(i) ***The strata where terminal cells are nonempty.*** *The pure odd-tensor terminal cells carry nonzero level ≤ 1 content only on inputs where the Mumford–Tate group of the relevant curve factors drops below the product symplectic group — i.e. on the* correspondence strata*: isogeny loci ($J(C_i) \sim J(C_j)$), endomorphism loci (*$\mathrm{End} \otimes \mathbb{Q} \supsetneq \mathbb{Q}$*: real/complex multiplication), and Weil-type loci.*

(ii) ***The isogeny-general strata are closed.*** *If the C_i are pairwise isogenous with* $\mathrm{End}(J)\otimes\mathbb{Q} = \mathbb{Q}$ *(no extra endomorphisms), the isogeny identifications identify the tensor content with the self-product analysis of Theorem S8(b): the terminal cells' content is the polarization channel, S5-closed.*

(iii) ***The high-endomorphism strata are the classical Weil sector.*** *On strata with* $\mathrm{End}(J)\otimes\mathbb{Q} \supsetneq \mathbb{Q}$*, the new level* ≤ 1 *sub-structures of the tensor tower are governed by the endomorphism algebra; at the extreme (CM/Weil type) they are exactly the* Weil classes *and their generalizations — the beyond-divisor sector of the Hodge conjecture for abelian varieties. The odd-tensor residual (R1) is thereby* merged *with the abelian residual (R2): a single named classical sector.*

[PROVEN HERE] *((i) is the contrapositive of the extremal-weight principle; (ii) reduces to S8(b) through the isogeny identifications; (iii) is the classification of level* ≤ 1 *sub-structures over endomorphism algebras — the territory the committed census worlds inhabit)*

Proof. (i) If MT is the full product GSp, Theorem S8's argument applies verbatim: sub-Hodge-structures are sub-representations, Schur calculus forces extremal constituents, no level ≤ 1 content beyond the polarization/volume channels. Nonzero terminal content requires a proper sub-representation structure, which for products of Jacobians is exactly a drop of the Mumford–Tate group; the MT-drop loci for curve products are the isogeny/endomorphism/Weil loci (the standard Noetherian stratification by Hom-lattices: the Hom-loci exhaust the MT-drops for products of Jacobians with H^1-content). (ii) A $\mathbb{Q}$-isogeny $\phi\colon J(C_i) \to J(C_j)$ identifies $H^1(C_j) \cong H^1(C_i)$ as rational Hodge structures; the tensor content becomes the self-product tensor $\otimes^n V$ with $V = H^1(C_i)$, and the level ≤ 1 analysis is S8(b)'s (the Schur calculus over GSp_{2g} depends only on the identified structure). With $\mathrm{End}\otimes\mathbb{Q} = \mathbb{Q}$ there are no further sub-reps. (iii) With $D := \mathrm{End}(J)\otimes\mathbb{Q} \supsetneq \mathbb{Q}$, the sub-representations of $\otimes^n H^1$ are governed by the commutant of D in the tensor algebra: the level ≤ 1 pieces are the D-invariant middle-typed tensors. At CM type (D a CM field) these are the CM-invariant middle tensors — the Weil-class generalizations; for Weil-type abelian varieties with associated elliptic curve E: the classical Weil classes, which on $E \times A$ land exactly in the balanced lattice (Round 33 Theorem S4's landing). The identification is functorial in the endomorphism algebra, so the merged sector is the classical beyond-divisor Weil sector. **[CERTIFICATE RECEIPT]** (V7: the CM-shadow computation — for the Gaussian letters the $D = \mathbb{Q}(i)$-invariant middle-typed tensors of $\otimes^3$ computed exactly: the middle part is D-rational, consistent with the S8 strata's $\theta \wedge V$-channel plus the Gaussian fold content; the hostile control — a non-CM random involution — fires.) □

5 Theorem S10: the re-drawn terminal map

Theorem 10 (The Strengthened Terminal Map, Round 34)**.** *On the open sector ($2 \leq p \leq \lfloor d/2 \rfloor$, $d \geq 4$), the rational Hodge conjecture is equivalent (Round 32 Theorem 4, unchanged) to [K-correspondences everywhere + transcendental algebraicity]. The input-axis map after this round:*

Sector / stratum	***Content***	***Status***	***Receipt***
Divisor algebra (all inputs)	*products of* $(1,1)$	*closed; DEAD as a balanced supply at every input*	*R33 S3*
Single-curve channels (all inputs reached)	$\delta \wedge V_* H^1(C)$	*closed, unconditional*	*R33 S5*
Twisted odd-tensor sectors	*tower images*	***closed — NEW (S7(i))***	*S7* ***[THIS SESSION'S OWN CALCULATION]***
Terminal cells, very-general distinct stratum	*pure tensor middles*	***EMPTY — NEW (S8a)***	*S8a* ***[THIS SESSION'S OWN CALCULATION]***
Terminal cells, self-product stratum	$\theta \wedge V$*-channels*	***closed, every genus — NEW (S8b)***	*S8b* ***[THIS SESSION'S OWN CALCULATION]***
Terminal cells, isogeny-general stratum	*identified tensor content*	***closed — NEW (S9ii)***	*S9* ***[THIS SESSION'S OWN CALCULATION]***
*Terminal cells, high-*End *strata*	D*-invariant middles = Weil-type content*	***merged into the classical Weil sector — NEW (S9iii)***	*S9*
Very-general curve 4-fold products, $(2,2)$ *on* $E \times X$	*divisor + balanced*	***fully closed — NEW***	*S8a(c)*
Abelian face	*balanced on* $E \times A$	*localized (R2)*	*R33 S4*
Projector shadow	π_{mid}	*bridged*	*R33 S2* ***[CONDITIONAL]***
Transcendental face	*fold-free core*	*irreducible*	*R32 Thm 4*

The named residual after this round: ***(R2′)*** *the merged beyond-divisor Weil/CM sector on* $E \times A$*-type inputs (the abelian face of S4 now carrying also the odd tower's high-*End *strata);* ***(R3)*** *the* K*-condition at general* (X, p) *beyond the channels;* ***(R4)*** *the transcendental fold-birth with the input-sensitivity gate.* ***The former (R1) is discharged as an independent residual:*** *its general strata are empty or closed, and its special strata are R2′'s territory.* Q1 *remains proven and certified; Q2 is not claimed; the closing string is not emitted.* ***[PROVEN HERE]*** *(as a map: each row's status is proven or honestly labeled)*

The quantifier line (§37)

What this round did to the universal quantifier: the odd-tensor family — which entered Round 33 as a residual family spread over *general* inputs — is now certified to carry irreducible content *only* on the classical correspondence strata (isogeny/CM/Weil), where it merges with the named abelian sector; on everything else it is tower-reducible (S7), empty (S8a), or closed (S8b, S9ii). An infinite family of open-sector inputs ($E\times$ very-general curve products at $(2,2)$) is now unconditionally closed. The Clay statement is not claimed; the closing string is not emitted.

6 The certificate

`tensor_tower_round34_checks.py` — bounded exact-arithmetic checks (this session's own computations, xorshift64* with seed $0x$C0FFEE123456789, a family and seed distinct from every prior certificate including Round 33's):

- **V1** the triple-tensor decomposition at every genus $2 \le g \le 6$: type-graded lattice models for $\otimes^3 V = \mathrm{Sym}^3 \oplus S_{(2,1)} \oplus \wedge^3$, the splitting $\wedge^3 = \wedge^3_0 \oplus [\theta \wedge V]$, dims $\binom{2g}{3}$, $2g$, $\binom{2g}{3} - 2g$, the level profile of every piece (only $\theta \wedge V$ at level 1), the $g = 2$ identity $\wedge^3 V = \theta \wedge V$;
- **V2** the outer-irreducibility, computed *exactly* by the period-monomial profiles: the 8 words of the g=1 triple tensor carry 8 distinct (k_1, k_2, k_3)-profiles, so at algebraically independent periods the middle-kernel equations are the identity — kernel exactly 0; at a shared transcendental period the kernel is exactly the 4-dimensional $S_{(2,1)}$-sum (verified against the Sym^3-complement by exact RREF: the two spans are disjoint and add to the full 8);
- **V3** the supply-transport shadow: on random finite correspondence models, the transport identity $\mathrm{cl}(Z_x) = e \otimes (\delta \wedge U_* \mu_1) + f \otimes (\delta \wedge U_* \mu_2)$ holds as a matrix identity, both evaluation orders, with the balanced-typing propagation verified;
- **V4** the base case: the finite shadow of the Lefschetz $(1,1)$ transport at curve carriers;
- **V5** the 5-letter type calculus at $g = 2$: the level profile of all Schur constituents of $\otimes^5 V$; only $\wedge^4 V \otimes V$ at level 1;
- **V6** the channel-typing shadow of Lemma 1 (the shift+twist type arithmetic on random graded models);
- **V7** the CM shadow: for the Gaussian letters ($D = \mathbb{Q}(i)$-module structure), the D-invariant middle-typed tensors of $\otimes^3$ computed exactly; the non-CM control fires;
- **V8** the NS-boundary at self-products: $[\wedge^3 V \cap (\mathrm{NS} \otimes V)] = \theta \wedge V$ by exact RREF at $g = 2, 3$;
- **V9** hostile controls: the sabotage battery — broken transport (a non-correspondence map), wrong twist (δ of one-off codimension), fake volume (a non-top exterior power), fake CM (a random involution) — every control must FIRE with the predicted failure.

All loops bounded before launch (enumerations $\le 2^{12}$, matrices $\le 12 \times 12$, instances ≤ 400, runtime < 60 s, zero floats, `Fraction`/int only), the vacuity scan (ledger-107 patterns) clean, deterministic double-run byte-identical.

7 The honest scope

Nothing in this round discharges (R2′), (R3) or (R4) at general input. What changed is the *trajectory*, for the second consecutive round: the residual family (R1) that looked like an unbounded general-input frontier is gone — replaced by three named objects with strictly smaller total content. The results that did the work are unconditional (S7, S8a on its stated stratum, S8b, S9i–ii) or classical-sector identifications (S9iii). The very-general closure of S8a(c) is a *stratum* result, stated with its stratum condition on the face (the MT condition is checkable, and the stratum is Zariski-dense); it is not a claim on the special strata, where the merged Weil sector lives and remains open. Per the campaign's banned substitutes discipline (PROMPT §3): curve-product strata results are

reported as *sector/stratum* closures with the residual named — never as the conjecture. The falsifiable line, sharpened: (R2′) one beyond-divisor balanced class on one $E \times A$ at a Weil-type input (the merged sector); (R3) the K-condition at one general input beyond the channels; (R4) the fold-birth at one transcendental input with the input-sensitivity gate firing.